HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2014)

Student No.: 03 Name: _____ Marks: / 34

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. If p and q are constants such that $px(x-1)+x^2 = qx(x-2)+4x$, then $p =$

   A. 1.

   B. 2.

   C. 3.

   D. 4.

(DSE-2014 PII Q03)

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2. If the angle and the radius of a sector are decreased by $x\%$ and 50% respectively so that its area is decreased by 90%, then $x =$

A. 20.

B. 40.

C. 60.

D. 80.

(DSE-2014 PII Q10)

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3. Let a_n be the nth term of a sequence. If $a_2 = 7$, $a_4 = 63$ and $a_{n+2} = a_{n+1} + a_n$ for any positive integer n, then $a_5 =$

A. 56.

B. 70.

C. 91.

D. 119.

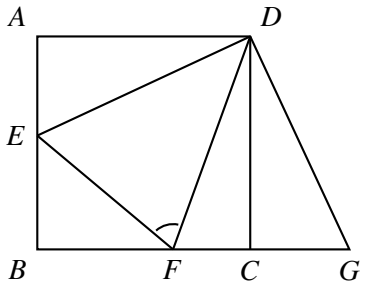
(DSE-2014 PII Q14)

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4. In the figure, $ABCD$ is a square. BC is produced to G such that $\angle CDG = 25°$. E is a point lying on AB such that $AE = CG$. If F is a point lying on BC such that $\angle CDF = 20°$, then $\angle DFE =$



A. 60°.

B. 65°.

C. 70°.

D. 73°.

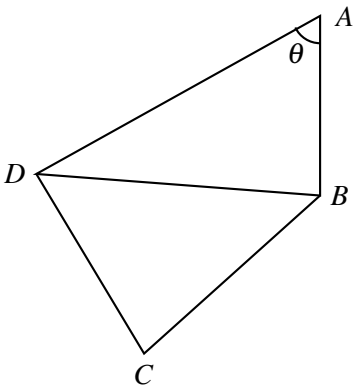
(DSE-2014 PII Q16)

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5. In the figure, $\angle ABD = \angle ADC = \angle BCD = 90°$. If $AB = \ell$, then $CD =$



A. $\ell \sin\theta$.

B. $\ell \cos\theta$.

C. $\ell \sin\theta \tan\theta$.

D. $\dfrac{\ell \tan\theta}{\cos\theta}$.

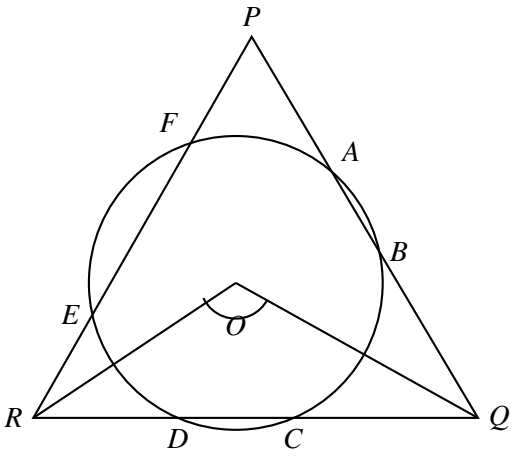
(DSE-2014 PII Q18)

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6. In the figure, O is the centre of the circle $ABCDEF$. $\triangle PQR$ intersects the circle at A, B, C, D, E and F. If $\angle QPR = 38^\circ$ and $AB = CD = EF$, then $\angle QOR =$



A. 109°.

B. 117°.

C. 123°.

D. 142°.

(DSE-2014 PII Q21)

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7. The rectangular coordinates of the point P are $(-1, \sqrt{3})$. If P is reflected with respect to the x-axis, then the polar coordinates of its image are

A. $(2, 210°)$.

B. $(2, 240°)$.

C. $(4, 210°)$.

D. $(4, 240°)$.

(DSE-2014 PII Q23)

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8. If a diameter of the circle $x^2+y^2-8x+ky-214=0$ passes through the point $(6,-5)$ and the slope of the diameter is -4, then $k=$

A. -6.

B. -4.

C. 13.

D. 70.

(DSE-2014 PII Q26)

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9. The stem-and-leaf diagram below shows the distribution of the ages of the passengers in a bus.

Stem (tens)	Leaf (units)
1	h 4 6
2	3 3 3 4 6 7 7
3	1 2 2 2 6 8
4	0 k

If the range of the above distribution is at least 33, which of the following must be true?

I. $0 \leq h \leq 3$

II. $3 \leq k \leq 9$

III. $3 \leq k - h \leq 5$

A. I only

B. II only

C. I and III only

D. II and III only

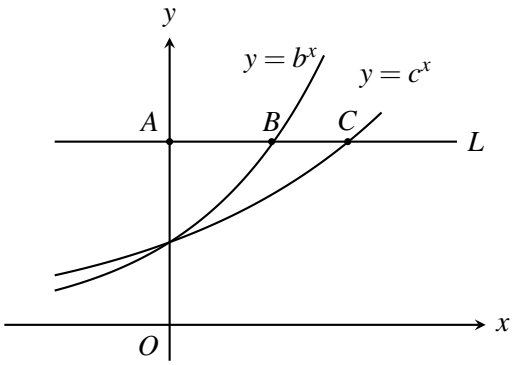
(DSE-2014 PII Q30)

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10. The figure shows the graph of $y = b^x$ and the graph of $y = c^x$ on the same rectangular coordinate system, where b and c are positive constants. If a horizontal line L cuts the y-axis, the graph of $y = b^x$ and the graph of $y = c^x$ at A, B and C respectively, which of the following are true?



I. $b < c$

II. $bc > 1$

III. $\dfrac{AB}{AC} = \log_b c$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2014 PII Q32)

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11. $7 \times 2^{10} + 2^8 + 5 \times 2^3 - 2^3 =$

A. 111010100000_2.

B. 111100010000_2.

C. 1110100100000_2.

D. 1111000010000_2.

(DSE-2014 PII Q34)

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12. If β is a real number, then $\dfrac{\beta^2+4}{\beta+2i} =$

A. $\beta-2i$.

B. $\beta+2i$.

C. $2-\beta i$.

D. $2+\beta i$.

(DSE-2014 PII Q36)

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13. If $m > 1$, which of the following are geometric sequences?

I. $2^m, 2^{2m}, 2^{3m}, 2^{4m}$

II. $m, 2m^2, 3m^4, 4m^8$

III. $\log m, \log m^2, \log m^4, \log m^8$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

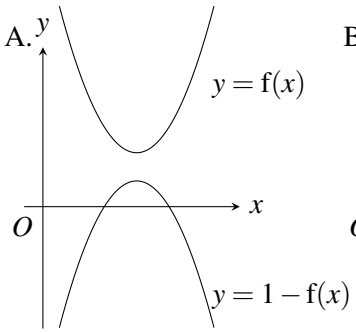
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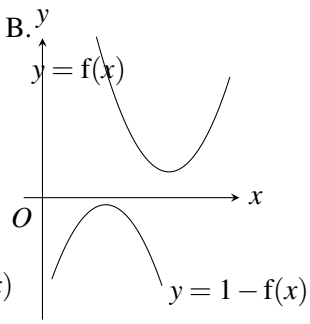
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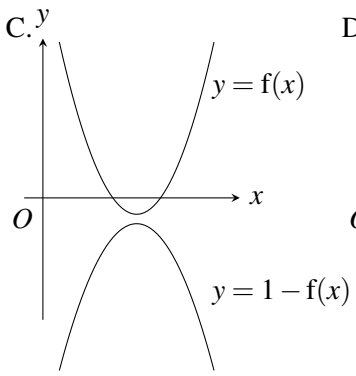
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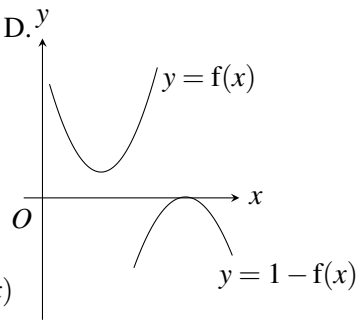
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14. Which of the following may represent the graph of $y = \mathrm{f}(x)$ and the graph of $y = 1 - \mathrm{f}(x)$ on the same rectangular coordinate system?

A.


B.


C.


D.


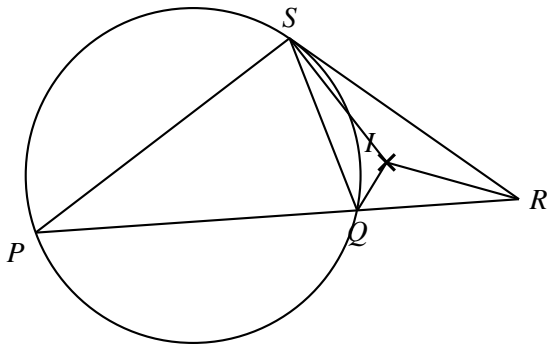
(DSE-2014 PII Q38)

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15. In the figure, PQS is a circle. PQ is produced to R such that RS is the tangent to the circle at S. I is the in-centre of $\triangle QRS$. If $\angle IRQ = 12^\circ$ and $\angle PSQ = 70^\circ$, then $\angle QPS =$



A. 24°.

B. 37°.

C. 43°.

D. 62°.

(DSE-2014 PII Q41)

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16. If the straight line $x - y = k$ and the circle $x^2 + y^2 + 2x - 4y - 1 = 0$ intersect at A and B, then the x-coordinate of the mid-point of AB is

A. $1 + k$.

B. $1 - k$.

C. $\frac{1+k}{2}$.

D. $\frac{1-k}{2}$.

(DSE-2014 PII Q42)

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17. If the variance of the four numbers a, b, c and d is 9, then the variance of the four numbers $14-a$, $14-b$, $14-c$ and $14-d$ is

A. 5.

B. 9.

C. 23.

D. 121.

(DSE-2014 PII Q45)

I can finish _____ questions

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